PEDRO HENRIQUE ROCHA DE ANDRADE

Computer Engineering
de Andrade, P. H. R.

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ACADEMIC SUMMARY

I'm currently a Computer Engineering student with a passion for astronomy and data science. I have experience in scientific research, with a focus on computational astrophysics, and a history of publications in national conferences. I'm always seeking to learn and contribute to the scientific community, exploring new technologies and methodologies to advance our understanding of the universe. I have a hobby in animations and simulations, which allows me to create engaging visualizations to communicate complex concepts in an accessible way, with a focus on science outreach of physics-related content.

ACADEMIC BACKGROUND

Bachelor in Computer Engineering

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

August. 2023 – present | Rio de Janeiro, Brazil

Technical Degree in Informatics

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

February. 2020 – December. 202 | Rio de Janeiro, Brazil

RESEARCH PROJECTS

Analysis of Dynamical Simulations of Merging Galaxy Clusters

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

September. 2022 – March. 2023 | Rio de Janeiro, Brazil

Junior Scientific Initiation Scholarship - CNPq
Supervisor: Dr. Ana Cecília Soja

MobFog at IFFMaker

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

August. 2023 – February. 2024 | Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - IFF
Supervisor: Dr. Ana Cecília Soja

Simulating the Impact of Satellites on Astronomical Observations

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

2024 – 2025 | Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja

Anomaly Detection in Stars of the Milky Way: Exploring Data from Gaia and Other Surveys with Machine Learning

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

2025 – present | Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja and Dr. Maria Luiza Linhares Dantas

EXTERNAL COURSES

Networks

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

August. 2020 | Rio de Janeiro, Brazil

Operational Systems

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

August. 2021 | Rio de Janeiro, Brazil

Server Administration

Fluminense Federal Institute *Campus* Bom Jesus do Itabapoana

August. 2022 | Rio de Janeiro, Brazil

RESEARCH INTERESTS

Stars | Milky Way | Galactic Archaeology | Star Evolution | Quantum Mechanics | Computational Physics | ML | AI | Physics | Astrophysics Computational | Deduction of Equations | Animations | Simulations

EVENTS

Feiras de Ciências

21st Brazilian Science and Engineering Fair
Poster – FEBRACE
Analysis of dynamic simulations of galaxy cluster mergers
2023 · São Paulo, Brazil

40th International Science and Technology Fair
Poster – MOSTRATEC
Analysis of dynamic simulations of galaxy cluster mergers
2023 · Novo Hamburgo, Brazil

IX Brazilian Science Initiation Fair
Poster – FEBIC
Simulating the Impact of Satellites on Astronomical Observations
2024 · Pomerode, Brazil

XVI Scientific and Technological Exhibition of the Açaí Institute
Poster – MCTIA
Simulating the Impact of Satellites on Astronomical Observations
2025 · Belém, Brazil

Congressos

44° CONUBES
Participant – CONUBES
2022 · Brasília, Brazil

XV Scientific and Technological Initiation Congress and IX Fluminense Congress of Postgraduate Studies
Oral Presentation – CONFICT
Simulating the Impact of Satellites on Astronomical Observations
2024 · Campos dos Goytacazes, Brazil

XVI Scientific and Technological Initiation Congress and X Fluminense Congress of Postgraduate Studies
Oral Presentation – CONFICT
Simulating the Impact of Satellites on Astronomical Observations
2025 · Campos dos Goytacazes, Brazil

XLVIII Annual Meeting of the Brazilian Astronomical Society
Poster – SAB
Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4
2025 · Caxambu, Brazil

XVII Scientific and Technological Initiation Congress and XI Fluminense Congress of Postgraduate Studies
Talk – CONFICT
Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4
2026 · Campos dos Goytacazes, Brazil

78ª Annual Meeting of Brazilian Society to Science Progress
Poster – SBPC
Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4
2026 · Niterói, Brazil

Winter School of Astrophysics
Poster
Unveiling the Solar vicinity: unsupervised mapping stellar populations with GCNS and GALAH DR4 using t-SNE
2026 · Rio de Janeiro, Brazil

XLIX Annual Meeting of the Brazilian Astronomical Society
Poster – SAB
How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE
2026 · Natal, Brazil

Teaching, Research and Extension Congress (CONEPE)

Presentation – CONEPE
LaTeX class: optimizing the formatting of academic papers.
2026 · Campos dos Goytacazes, Brazil

Semanas Acadêmicas

VIII Knowledge Showcase and I Job Fair
Participant
2020 · Bom Jesus do Itabapoana, Brazil

IX Knowledge Showcase and II Job Fair
Participant
2021 · Bom Jesus do Itabapoana, Brazil

I Computational Engineering and Management Week
Participant
2021 · Bom Jesus do Itabapoana, Brazil

X Knowledge Showcase and III Job Fair
Speaker
Analysis of dynamic simulations of galaxy cluster mergers
2022 · Bom Jesus do Itabapoana, Brazil

II Computational Engineering and Management Week
Participant
2022 · Bom Jesus do Itabapoana, Brazil

XI Knowledge Showcase and IV Job Fair
Participant
MobFog no IFFMaker
2023 · Bom Jesus do Itabapoana, Brazil

III Computational Engineering and Management Week
Participant
MobFog no IFFMaker
2023 · Bom Jesus do Itabapoana, Brazil

XII Knowledge Showcase and V Job Fair
Participant
2024 · Bom Jesus do Itabapoana, Brazil

IV Computational Engineering and Management Week
Speaker and Organizer
2024 · Bom Jesus do Itabapoana, Brazil

PUBLICATIONS

Link Nasa ADS: NASA ADS - Pedro H. R. de Andrade

Proceedings

[1] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *Boletim da Sociedade Astronômica Brasileira*. Vol. 37. 1. 2026, pp. 308–309.

[2] Pedro Henrique Rocha de Andrade, Ana Cecilia Soja, and Maria Luiza Linhares Dantas. « How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE ». in: *Boletim da Sociedade Astronômica Brasileira*. Natal, RN, 2026.

[3] Pedro Henrique Rocha de Andrade, Ana Cecilia Soja, and Maria Luiza Linhares Dantas. « Propriedades estelares e características químicas do Gaia Catalogue of Nearby Stars observado pelo GALAH DR4 ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2026.

[4] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Anais da 78ª Reunião Anual da Sociedade Brasileira para o Progresso da Ciência (SBPC)*. Niterói, RJ: UFF, 2026.

[5] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Resumos da Escola de Inverno de Astrofísica do Observatório Nacional*. Rio de Janeiro, RJ: Observatório Nacional, 2026.

[6] Maycon Jorge Deláqua da Silva, Pedro Henrique Rocha de Andrade, and Ana Cecilia Soja. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2025.

[7] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *XVI Mostra de Ciência e Tecnologia do Instituto Ação (MCTIA)*. 2025.

[8] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e IX Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2024.

[9] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais da IX FEBIC – Feira Brasileira de Iniciação Científica*. 2024. ISBN: 978-65-991144-6-5.

[10] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Análise de simulações dinâmicas de aglomerados de galáxias em fusão ». In: *Anais da 21ª FEBRACE – Feira Brasileira de Ciências e Engenharia*. Vol. 21. 2023, p. 5. ISBN: 978-65-89190-18-9.

[11] Gabriel Coelho de Souza Poubel et al. « MobFog no IFFMaker ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 21.

Articles

[12] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *arXiv e-prints* (Dec. 2025). DOI: 10.48550/arXiv.2512.10037.

AWARDS

IX FEBIC
3rd Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"

XVI MCTIA
1st Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"

STATISTICS

3 Oral Presentations
National Events

7 Posters
National Events

9 Science Fairs/Congresses
National Events

10 Academic Weeks
Regional Events

PROGRAMMING

Python | git Git | Github | LaTeX | C++ | Java | NodeJS

LANGUAGES

Portuguese	●●●●●
English	●●●●○
French	●○○○○
Spanish	●●○○○